01 / PROBLEM AND CONTRIBUTION

# CMOS constant-gm current references

EEE 523 · Advanced Analog IC Design | Spring 2026 | Individual design

**RESULT AT A GLANCE**

Designed three reference topologies; Design 1 produced approximately 10.1 µA at 2.5 V and a reported 4.8% endpoint gm variation over -20 to 85 °C.

## The engineering question

How can a beta-multiplier reference provide useful bias current and relatively stable transconductance across temperature?

## Situation

Analog stages need a usable bias reference as supply and temperature change. This study compares three CMOS reference circuits rather than relying only on a nominal bias point.

## Task

Choose device sizing and resistors, include start-up circuitry, and compare branch currents and transconductance across the specified sweeps.

## My contribution

Designed and characterized three CMOS reference circuits, including start-up devices.

## Tools and concepts

CMOS / Current mirrors / Beta multiplier / Cadence Virtuoso / Temperature analysis

02 / DESIGN AND METHOD

# From the task to implementation

### 01 / Implementation step

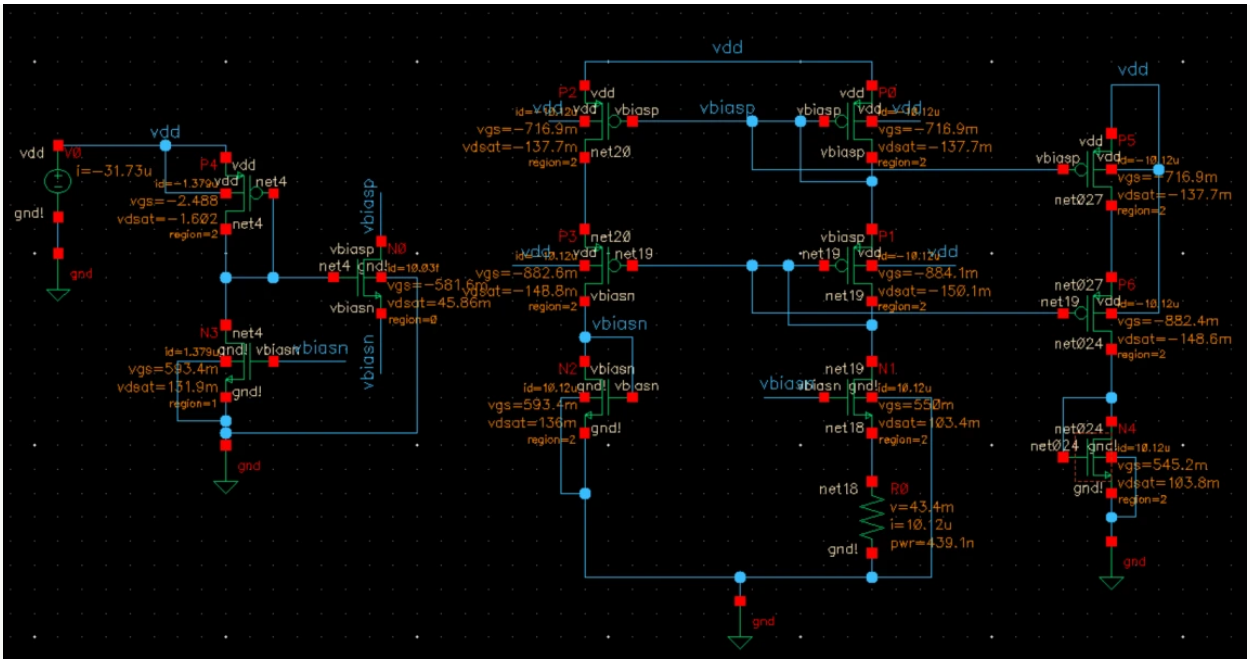
Compared three reference topologies in the TSMC 0.25 µm CMOS design library. Selected resistor values and device sizes and tabulated nominal operating points.

### 02 / Implementation step

Swept the supply from 2 to 3 V to compare the two branch currents and their mismatch. Used a diode-connected NMOS load to study transconductance from -20 to 85 °C.

### Reference design choices

Device ratios and the current-setting resistor determine the bias point. Start-up devices are included in the reported designs; a nominal current result alone does not establish reliable start-up under every possible supply ramp.




**Bias circuit schematic**

One of the bias topologies evaluated in the constant-gm study.

03 / RESULTS AND ENGINEERING PERSPECTIVE

# Evidence, interpretation, and next steps

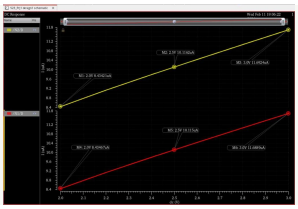
| | |
|---|---|
| **Nominal supply** | 2.5 V |
| **Design 1 current** | ≈10.1 µA |
| **Temperature sweep** | -20 to 85 °C |
| **Design 1 gm variation** | 4.8% |

## Reported result

For Design 1, the nominal branch currents are approximately 10.116 and 10.115 µA at 2.5 V. The report calculates a 4.8% endpoint gm variation normalized to the 27 °C value.

## Comparing supply and temperature response

The supply sweep compares reference-branch currents, while the diode-connected NMOS load provides the temperature-dependent transconductance observation. The reported 4.8% endpoint change is normalized to the 27 °C value for Design 1.



**Bias sweep results**

Simulation plot reproduced from the bias-design report.

## Scope and limitations

The 4.8% number belongs specifically to Design 1 and the stated temperature sweep. It is not a statistical matching or full PVT result.

### Possible next steps - proposed, not completed results

• Summarize all three topologies using the same supply, temperature, and normalization conventions.

• Add explicit start-up transients and process-corner checks before describing the circuit as robust across PVT.

Source material: Project1_AAIC.pdf, pp. 3-8; subsequent sections compare Designs 2 and 3.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.